

Real-Time Machine Learning-Based Threat Detection for ISP Network Traffic

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Abstract

1 Introduction

Over the past decades, the Internet and the number of connected IoT devices have experienced an exponential increase. With this increase, many vulnerabilities and weaknesses were induced, making the cybersecurity field a paramount priority. By tailoring social connections, developing public opinions, and shaping economic balance, the network plays a vital role in human life.

Due to the complexity of the software and the evolution of cyber threats, it is necessary to acquire strong mechanisms for recognising, comprehending, and mitigating vulnerabilities. Firewalls and IDSes represent defense strategies that help in monitoring events to provide warnings or alerts about unusual traffic. To enhance the ability of IDSes to detect novel or previously unseen attacks, unsupervised learning machine learning models have gained popularity, particularly because they don't require labeled data and they could identify abnormal and unusual patterns in the network flow.

Researchers have extensively explored machine learning (ML) techniques, and have adopted various detection methodologies. The main type of IDSes are, in general: model-based, signature-based, unsupervised and hybrid. Each of these detection approaches has drawbacks, including the necessity of human interaction for data tagging, the inability to handle invisible attacks, and the necessity of frequent updates because the network traffic is changing always.

Additional Keywords and Phrases: IDS(Intrusion Detection Systems), anomaly detection, supervised learning, unsupervised learning, active learning, BERT, decision tree-based algorithms

1.1 Problem Statement

Bright Access is an internet service provider with its own proprietary network infrastructure. Leveraging in-house developed technologies, Bright Access enables the configuration, monitoring, and management of telephony, data communication, and fiber-optic networks. Partell is currently developing a system capable of real-time classification of network packets. This system will identify potentially malicious traffic by analyzing network behavior. Traditional detection methods rely on fixed signatures or pattern recognition after incidents occur, which introduces latency and reduces effectiveness. Partell seeks to improve upon these methods using machine learning (ML) models for proactive, intelligent threat detection.

This research has the goal of creating a transition from a passive conduit to an active guardian of internet security. It aims to investigate the trade-offs between different ML models in terms of accuracy and time efficiency and to create an innovative system that performs threat detection on ISP-level traffic.

Based on the problem statement, the following research questions were addressed.

1.2 Research Questions

RQ1: How do various supervised-learning algorithms interact with various query strategies to result in the highest performance and labeling efficiency in an active learning framework?

RQ2: How do various clustering algorithms perform in binary classification and outlier detection tasks?

Section 2 introduces a literature overview of the relevant studies related to this research study, Section 3 presents the methodological approach applied during this research. The findings of the research study are presented in Section 4 and discussed in Section 5. Sections 6 and 7 present the design overview and the technical details of the prototype. The research paper ends with a conclusion in Section 8 and future research recommendations in Section 9.

2 Literature Review

There are many distinct types of intrusion detection systems, and they can be distinguished by their operational style. Nowadays, signature-based IDSes like Suricata [1] and SNORT [2] make up the majority of those utilized in production. With these solutions, security professionals are updating the IDS rules by create a signature for every new attack that is taken place. Consequently, only when an activity exhibits a pattern that corresponds to a known attack signature is it marked as abnormal.

Because of the shortcomings of signature-based intrusion detection systems, researchers have looked into more sophisticated methods, including machine learning.

These strategies rely on supervised learning models. Emad E. Abdallah et al.[] explores the application of supervised learning in respect with IDSes. This category of IDSes use a supervised-learning model which has contain a knowledge base constructed on labeled data. The new activity is then classified into malicious or normal based on the prediction of the trained model. Typical supervised-learning models used in this approaches are Logistic Regressor, decision-tree algorithms, random forest, (deep) neural networks architectures.

Anomaly-based IDSes are systems which rely on unsupervised learning principle by identifying anomalies/outliers by learning the normal conduct of the network traffic. Binary classification using clustering and Outlier detection are 2 unsupervised approaches present in this case study. Common clustering algorithms used are K-means and DBSCAN. Typical outlier detection framework are represented SVM(support vector machines) and KNN(Kth-Nearest-Neighbour) algorithms. Table 1 presents an overview of some existing solutions.

2.1 Limitations

One limitation may be induced by the lack of real labeled data. The majority of datasets represent simulations of different attacks. This has a negative impact over detecting real-world attacks. Moreover, this negative impact may be crucial for model's training phase, leading to a packet misclassification. Another limitation is related to the signature-based IDSes. They work based on a set of implicit rules. Without expert's intervention, signature-based IDSes cannot identify new patterns in the network traffic. Whereas unsupervised learning present advantages in terms of identifying new attacks, their primary limitation is the high rate of false positives. Any deviation from the normal data is classified as a potential attack.

3 Methodology

This section outlines the methodological approach used during this research. This includes the following:

1. *Data Collection:* Collecting and testing labeled datasets for training, testing, and validating the active-learning model's predictions
2. *Feature Selection:* Select the appropriate features that contribute the most to the prediction target

Paper	Proposed solution	Description of the approach
Boukela et al.(2021)[3]	Hybrid IDS using Active Learning with DNN(supervised module) and KNN(unsupervised module)	The IDS processes the network traffic in an incremental procedure using sliding window of certain size. It uses an active learning model constructed on a DNN (supervised module) architecture for identifying known pattern and KNN (unsupervised module) model for identifying unknown patterns. Through uncertainty sample query function, the AL model asks an expert to verify the label of the most uncertain predictions.
Amado et al.(2023)[4]	Malicious traffic detector using Kitsune’s KitNET neural network(unsupervised module)	The solution uses Perforce(P4) software to extract packets over the network switch. The neural network is built on autoencoders architecture. The performance of the solution is measured using the AUC metric. The threshold for considering a packet malicious is less than 0.5.
Nico Görnitz et al.(2009)[5]	ActiveSVDD: anomaly detection using active learning task using Support Vector Data Description	This approach uses hypersphere-based anomaly detection(semi-supervised learning). It combines 2 strategies: <i>Marginal Strategy</i> : implies examination of points closed to the decision hypersphere and <i>Exploration strategy</i> : determine attacks by exploring unknown areas in the feature space.
Paolini et al.(2024)[6]	Packet Classification Prototype using CNN-architecture	This approach collects every specific feature related to an individual packet and creates an matrix of size $n \times p$ where n is the number of features selected and p is the number of packets within a single flow. It uses a custom CNN architecture(supervised-learning module) for training and testing. It measures the performance of the model using the accuracy metric.

Table 1: Summary of existing solutions